

Route 1 — NL

PROMPT

Solve the algorithmic problem step by step in natural language.
Return JSON: {simulation, answer}.

CONSTRAINT

no code

EXAMPLE OUTPUT • Q: 123 + 456

```
{
  "simulation":
    "3+6=9, 2+5=7, 1+4=5",
  "answer": "579"
}
```

answer

579

Route 2 — Code + sim.

PROMPT

Solve the algorithmic problem.
Return code + an NL simulation.
Return JSON: {code, simulation, answer}.

CONSTRAINT

same JSON schema

EXAMPLE OUTPUT • Q: 123 + 456

```
{
  "code": "def solution():
    return 123+456",
  "simulation": "adds",
  "answer": "579"
}
```

answer

579

Route 3 — Code + Python

PROCEDURE

1. Extract solution() from Route 2.
2. Run in sandboxed Python (5 s).
3. Compare output to gold.

NOTE

no additional LLM call

EXECUTION • CODE FROM ROUTE 2

```
>>> def solution():
...     return 123 + 456
>>> solution()
579
```

answer

579